

Report I — Step 1: Understanding the Challenge

Course: MAS DSE 260A (Capstone Project)

Project Title: Daily Electric Load Forecasting

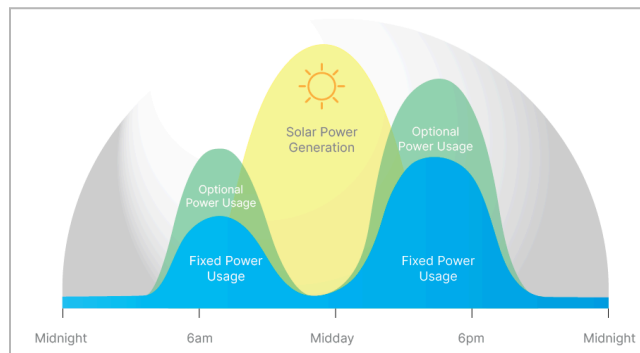
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Advisor(s): Prof. Raymond de Callafon (callafon@ucsd.edu)

1. Motivation

Planning and controlling electricity consumption is increasingly important for residential customers, commercial entities, and critical infrastructure, especially when paired with distributed energy resources like PV (photovoltaic) power generation and battery storage. With an accurate forecast of power consumption (load), one can reduce electricity costs through two widely used strategies:

- **Peak shaving:** Reducing electricity use during the most expensive and demanding times (peaks) by using stored energy (batteries/solar), lowering consumptions, or other approaches.
- **Load shifting:** Moving electricity consumption from high-demand (peak) times to low-demand (off peak) times to take advantage of cheaper electricity rates.



2. Challenge

The main **challenge is forecasting daily load consumption accurately enough to support energy optimization and scheduling decisions, including peak shaving and load shifting.** High variability and strong temporal dependencies – such as operational patterns, weather effects, or human activity – can make short-term forecasting difficult, leading to suboptimal battery usage, reduced peak-shaving effectiveness, and higher energy costs.

Battery control and optimization frameworks rely heavily on accurate load forecasts; poor forecasting directly degrades optimization performance. Therefore, this capstone focuses on

developing an algorithm to predict daily load profiles with sufficient accuracy to support downstream scheduling and storage control decisions. The project is motivated by a real industrial application in which a commercial entity integrating PV generation and battery storage seeks to understand how reliably daily load can be predicted to enable efficient battery scheduling and reuse, potentially in coordination with PV production.

3. Opportunities as a Set of Questions

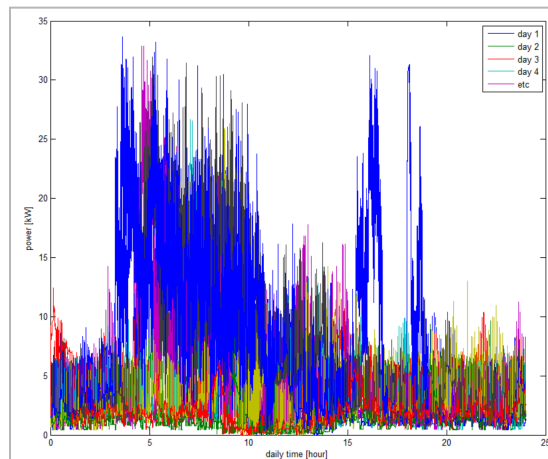
The project's opportunities can be framed as specific, data-driven questions:

1. How accurately can we predict a full-day electric load profile using historical daily load time-series data? Which metric(s) will be used to quantify the quality of the prediction (accuracy)
2. Which modeling approaches best capture temporal patterns in load consumption?
3. How does temporal resolution (e.g., 1-second, 1-minute, or 15-minute data) impact forecast accuracy and practical usability for energy optimization?
4. How robust are load forecasting models to noise, data quality issues, and irregular or unexpected load behavior?
5. Can we implement a recursive forecast that updates predictions as new measurements arrive (e.g., every 5–10 minutes), and does this reduce error compared to a once-per-day forecast?

4. Data Sources

Primary Dataset

The primary dataset consists of high-resolution (1-second) daily electric load time-series data from a commercial facility planning to integrate photovoltaic (PV) generation and battery storage. The data is organized as multiple daily load profiles, with each day containing 24 hours of power consumption measurements (in watts). The dataset exhibits stable baseline loads, cyclic operational patterns, and occasional event-driven spikes, with clear differences between working days and non-working days or holidays.



Supporting Information

Basic metadata, including day indices and calendar information, is available to distinguish workdays from non-workdays and to support model training and cross-validation.

Data Limitations

The dataset reflects real-world operating conditions and includes noise, occasional missing or invalid measurements due to sensor or meter issues, and timing variability in daily operations. External variables such as weather are not guaranteed to be available, requiring forecasting approaches to primarily rely on historical load behavior.

Data Assumptions

- Real-power measurements with a specific accuracy.
- Expected number of data points is $24 \times 60 \times 60$ per day. Missing data points are represented as NaN values in the dataset.
- Real-power variations are assumed to be driven solely by commercial activity and are considered uncorrelated with weather conditions, outdoor temperature, or humidity.
- A categorical distinction between workdays and non-workdays is provided in the dataset.

4. Approach

Our approach follows an end-to-end forecasting workflow:

Preprocessing → EDA → Baseline Modeling → Advanced Modeling → Evaluation

4.1 Preprocessing

During preprocessing, we will convert raw data into clean daily profiles suitable for modeling.

- **Load and validate structure:** confirm daily matrix shape and date mapping.
- **Unit normalization:** represent load as kW consistently
- **Missing values and anomalies:**
 - Detect suspicious zeros or dropouts (e.g., abrupt jumps to zero and back).
 - Treat true zeros carefully (e.g., facility shutdown) vs missing readings.
 - Impute short dropouts (forward-fill / interpolation) or mask them for models that can handle missingness.
- **Transformations:** Add or transform features, as needed, for modeling.
 - Time features: season, quarter, month, day of the month, day of the week, time of the day, hour interval, etc.
 - Working / Non-working day flags

4.2 Exploratory Data Analysis (EDA)

Explore the data to identify any trends or correlations.

- **Daily overlays** to see variability across days and identify typical patterns.
- **Clustering / grouping** to distinguish working vs non-working days based on total energy, peak load, or profile similarity.
- **Event pattern identification:** recurring cycles (e.g., refrigeration) and intermittent high-power events (e.g., oven spikes).
- **Timing variability:** quantify time shifts of peaks across days.

4.3 Baseline Modeling

Develop baseline forecast models to better understand the prediction of the data.

- **Average daily profile baseline:** Compute mean load profile across historical workdays (and separately across non-workdays).
- **Persistence / “yesterday equals today” baseline:** Predict today’s profile using the previous day (or most similar recent day).
- Persistence due to workday non-workday distinction.
- **Simple feature baselines:** Use daily summary features (total energy, peak load, time-of-peak) to choose a representative template day.

4.4 Advanced Modeling

We will explore multiple complementary modeling strategies for daily load forecasting:

- **PCA-Based Forecasting:** Use PCA/SVD to capture dominant daily load patterns by representing each day as a linear combination of a small number of principal components. Simple alignment strategies or time-shift–robust feature representations will be considered to reduce smearing effects. Forecasting is performed by predicting component weights for a target day using historical data and partial-day observations.
- **Neural Network Forecasting:** Apply neural time-series models (e.g., temporal convolution or RNN/LSTM-based architectures) to learn nonlinear temporal dependencies in load data. Models will be designed to condition on partial-day observations and calendar-derived features while relying primarily on historical load behavior.
- **Recursive / Rolling Forecasting:** Implement a recursive forecasting framework in which predictions are updated as new measurements arrive (e.g., every 5–10 minutes). This includes short-horizon rolling forecasts and rest-of-day predictions that evolve throughout the day, enabling realistic operational use for energy optimization.

4.5 Evaluation

Identify primary success metrics:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)

- Peak error (difference between predicted and actual daily max)
- Energy error (difference in daily total energy usage)
- Time-Weighted Least Square measure to account for electricity pricing and/or importance of forecast

We will evaluate:

- Overall performance across all days
- Stratified performance: workdays vs non-workdays
- Robustness to missing data
- Performance under rolling forecast updates

Finally, we will perform cross-validation by withholding entire days from training to test generalization, aligned with the required deliverable of predicting load data not included in the dataset.

5. Team Roles and Responsibilities

Role	Responsibility	Team Member(s)
Project Manager	Timeline tracking, coordination, and communication	Spencer
Budget Manager	Manage the project budget	Sean
Data Engineering	Data ingestion, cleaning, and preprocessing	Frank
Data Analyst	Exploratory data analysis and visualization	Sean
Model Engineer	Development, training, and evaluation of forecasting models using appropriate performance metrics and cross-validation.	Frank
Record Keeper/ Project Coordinator	Organize project documentation, resources, and code	Spencer
Technical Writer	Writing the reports, organizing figures and results, and ensuring clarity and consistency	Sean, Spencer, Frank

6. Project Coordination and Communication Plan

Meetings

- **Weekly advisor check-in (Mondays):** progress updates, technical questions, and next-step alignment.
- **Weekly team work session:** implementation sync, unblock tasks, and merge progress.
- **Additional ad-hoc meetings** as needed before report deadlines.

Tools

- **GitHub** for version control and code review.
- **Google Drive** for shared writing, figures, and references.
- **Discord** for daily communication and quick questions.
- **Zoom** for meetings and screenshare debugging.

Workflow

- Maintain a shared task board (GitHub Projects) with:
 - backlog, in-progress, review, done.
- Define milestones:
 - Step 1: Understanding the Challenge
 - Step 2: Designing the Data Acquisition and Preparation Pipelines
 - Step 3: Exploring Data
 - Step 4: Defining Your Hypothesis and Minimum Viable Modeling Product
 - Step 5: Creating a Solution Architecture for Modeling and Optimization
- Keep a running experiment log documenting:
 - dataset version, preprocessing settings
 - model hyperparameters and evaluation metrics
 - plots for qualitative comparison

7. Individual Contributions in Step 1

Sean He

- Drafted Report I.
- Consolidated project background and clarified deliverables.
- Outlined dataset constraints and data-quality issues based on the project meeting

Spencer Hoyle

- Reviewed and updated Report I
- Defined Team Roles and Responsibilities
- Created Discord channel and Github repository for team communication

Frank Chavezsosa

- Reviewed and updated Report I
- Contributed to brainstorming project ideas
- Conducted background research by identifying and providing relevant research papers